LESSON 3 · HANDS-ON MRI #1

The Scan Converter

Getting Hands-On with MRI #1: Converting DICOM to NIfTI

TOOLS UNLOCKED

MRIcroGL · dcm2niix · .dcm · .nii.gz · .json

From the scanner to a file you can analyze



1 · SCAN

The MRI scanner saves each slice as its own file

↓ raw_dicoms/sub-001/



2 · DICOM

A folder of slices per scan

FLAIR_DICOM/ · MPRAGE_DICOM/

↓ click: MRIcroGL or type: dcm2niix



3 · NIfTI

One 3D brain volume, plus its scan details

.nii.gz + .json

1 Open the subject: two scans inside

raw_dicoms/sub-001

```
lindseyko@Lindseys-MacBook-Air raw_dicoms % cd sub-001
lindseyko@Lindseys-MacBook-Air sub-001 % ls
FLAIR_DICOM     MPRAGE_DICOM
```

2 Each scan is a folder of slice files

FLAIR_DICOM

```
lindseyko@Lindseys-MacBook-Air FLAIR_DICOM % ls
1.2.840.113619.6.495.64968897418720824957614882386711220946-4-1-z43833.dcm
1.2.840.113619.6.495.64968897418720824957614882386711220946-4-10-z43836.dcm
1.2.840.113619.6.495.64968897418720824957614882386711220946-4-11-z4384r.dcm
1.2.840.113619.6.495.64968897418720824957614882386711220946-4-12-z4385r.dcm
```

3 Spot the trap: 1, 10, 100 … not slice order

MPRAGE_DICOM

```
lindseyko@Lindseys-MacBook-Air MPRAGE_DICOM % ls
1.2.840.113619.6.495.64968897418720824957614882386711220946-3-1-z4359o.dcm
1.2.840.113619.6.495.64968897418720824957614882386711220946-3-10-z435ai.dcm
1.2.840.113619.6.495.64968897418720824957614882386711220946-3-100-z4360c.dcm
```

dcm2niix reads the true order from the file headers